Serina Chang

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Most Recent CV Here

ACADEMIC POSITIONS

University of California, Berkeley, Berkeley, CA — 2025-present
Assistant Professor, Department of Electrical Engineering and Computer Sciences (50%), UCSF UC Berkeley Joint Program in Computational Precision Health (50%)
Member of the Berkeley AI Research (BAIR) Lab and the Center for Human-Compatible AI (CHAI)

Microsoft Research, New York, NY — 2024-2025
Postdoctoral Researcher, Computational Social Science group

EDUCATION

Stanford University, Stanford, CA — 2019-2024
Ph.D., Computer Science
Advisors: Jure Leskovec, Johan Ugander; *Thesis committee*: Dan Jurafsky, Eric Horvitz
Dissertation: "Computational Methods for Human Networks and High-Stakes Decisions"

Columbia University, New York, NY — 2015-2019
B.A., Computer Science (major), Sociology (concentration), *magna cum laude*
Advisor: Kathleen McKeown

HONORS

Forbes 30 under 30 (United States, Healthcare)	2026
UC Berkeley, College of Computing, Data Science, and Society, Machine Learning Research Award	2026
ACM SIGKDD Dissertation Award	2025
Google Research Scholar Award	2025
Rising Stars in Data Science, University of Chicago and University of California San Diego	2023
EECS Rising Stars, Georgia Institute of Technology	2023
Future Faculty Symposium Scholar, Cornell University	2023
Meta PhD Fellowship, Computational Social Science	2022
ACM SIGKDD 2021 Best Paper Award, Applied Data Science Track	2021
Graduate Research Fellowship, National Science Foundation	2019
The Finch Family Fellowship, Stanford University, School of Engineering	2019
Outstanding Undergraduate Researcher Award, Computing Research Association	2019
Phi Beta Kappa, Columbia University	2019
Theodore R. Bashkow Award, Columbia University, Computer Science	2019
Academic Excellence Award, Columbia University, Computer Science	2019
King's Crown Leadership Award, Innovation and Enhancement, Columbia University	2018
Dean's List (all semesters), Columbia University	2015-2019

RESEARCH INTERESTS

My research lies at the intersection of **AI and human behavior**, including developing human-centered AI models, evaluating human-AI interactions, simulating and inferring human behaviors with AI, and modeling population behaviors to guide high-stakes public health and policy decisions.

PAPERS

*indicates co-first authorship.

Under Review / Preprint

[34] Quantifying the Utility of User Simulators for Building Collaborative LLM Assistants
Joseph Suh, Ayush Raj, Minwoo Kang, and **Serina Chang**

[33] Dataset Signatures in Human-LLM Interactions and User Modeling
Joseph Suh and **Serina Chang**

[32] Learning Demographic-Conditioned Mobility Trajectories with Aggregate Supervision
Jessie Li, Zhiqing Hong, Toru Shirakawa, Shashank Mehrotra, and **Serina Chang**

[31] Whose Story Does AI Tell? Investigating LLMs as News Agents
Samia Menon*, Kanav Mittal*, Abigail O'Neill, and **Serina Chang**

[30] Neyman Orthogonal Activation Steering
David Arbour, Erfan Jahanparast, **Serina Chang**, and Avi Feller

[29] ORQA: An Occupation-Realistic Question and Answer Framework for LLM Professional Knowledge
Shreyas Krishnan, **Serina Chang**, and Abhishek Nagaraj

[28] PreciseCoder: Faithful and Generalizable Debugging Beyond Unit-Test Rewards
Yunkai Zhan, Miaosen Chai, Shangshang Wang, Jike Zhong, Yuqing Yang, Deqing Fu, **Serina Chang**, and Wang Bill Zhu

[27] Evaluating Large Language Models for Guiding Policy-Level, Multi-Objective Decisions
Ritviksiddha Penchala, **Serina Chang**, and Reza Yaesoubi

[26] Behavioral AI: Building Algorithms That Understand Us
Keyon Vafa, Ilia Sucholutsky, Katherine M. Collins, Michael S. Bernstein, Andreea Bobu, **Serina Chang**, Diag Davenport, Kawin Ethayarajh, Katy Ilonka Gero, Thomas L. Griffiths, Hoda Heidari, Hannah Rose Kirk, Jon Kleinberg, and Sendhil Mullainathan

[25] Modelling the Public-Health Impact of Indoor Air Quality Interventions on Respiratory Virus Transmission
Adam Howes, Geetha Jeyapragasan, Richard Williamson, David Carel, Harrison Koos, Jacob L. Swett, James Montavon, Vivian Belenky, Paul Lietar, Richard Fitzjohn, Giovanni Charles, **Serina Chang**, Thomas Brewer, and Charles Whittaker

Published / Accepted

[24] Same Event, Different Truths: How Next-Generation Social Media Platforms Reframe Political Events
Mao Li, Xinyi Chen, and **Serina Chang**
ICWSM 2027

[23] Political Neutrality as Balanced Approval: A Large-Scale Human Evaluation of AI Responses
Jonathan Stray*, David Zhai Yang*, Steven Luo*, Miu Nicole Takagi, and **Serina Chang**
NeurIPS 2026, Evaluations and Datasets Track
Spotlight (33 Spotlight + 15 Oral out of 3757 submissions, 1.3%)

[22] Graph-Based Alternatives to LLMs for Human Simulation
Joseph Suh, Suhong Moon, and **Serina Chang**
ACL 2026 (main, long paper)

[21] Valid Survey Simulations with Limited Human Data: The Roles of Prompting, Fine-Tuning, and Rectification
Stefan Krsteski, Giuseppe Russo, **Serina Chang**, Robert West, and Kristina Gligorić
ACL 2026 (main, long paper)

[20] What Do Large Language Models Know About Opinions?
Erfan Jahanparast, Zhiqing Hong, and **Serina Chang**
ICLR 2026

[19] Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation
Song-Ze Yu, Joseph Suh, **Serina Chang**, and David M. Chan
EMNLP 2026, System Demonstrations Track

[18] Global Approaches to Infectious Disease Surveillance and Modelling
Mark Khurana, Joseph Tsui, Bernardo Gutierrez, Ayush Chopra, Neil Scheidwasser, Harrison Zhu, **Serina Chang**, David Duchêne, Cathal Mills, Rhys Inward, ..., Samir Bhatt, Moritz Kraemer
Nature Medicine, 2026

[17] Language Model Fine-Tuning on Scaled Survey Data for Predicting Distributions of Public Opinions
Joseph Suh*, Erfan Jahanparast*, Suhong Moon*, Minwoo Kang*, and **Serina Chang**
ACL 2025 (main, long paper)
Also presented at American Association for Public Opinion Research (AAPOR) Conference 2025 (oral) and the Social Sim and NLPOR Workshops at COLM 2025

[16] ChatBench: From Static Benchmarks to Human-AI Evaluation
Serina Chang, Ashton Anderson, and Jake Hofman
ACL 2025 (main, long paper)

[15] LLMs generate structurally realistic social networks but overestimate political homophily
Serina Chang*, Alicja Chaszczewicz*, Emma Wang, Maya Josifovska, Emma Pierson, and Jure Leskovec
ICWSM 2025
Also presented at IC^2S^2 2024 as **Plenary Talk** (2.8% of submissions)

[14] Learning production functions for supply chains with graph neural networks
Serina Chang, Zhiyin Lin, Benjamin Yan, Swapnil Bembde, Qi Xiu, Chi Heem Wong, Yu Qin, Frank Kloster, Xi Luo, Raj Palleti, and Jure Leskovec
AAAI 2025, AI for Social Impact Track
Oral presentation (4.9% of submissions)
Also presented at Stanford Graph Learning Workshop 2023 and Stanford Causal Science Conference 2023

[13] Artificial intelligence for modelling infectious disease epidemics
Moritz U. G. Kraemer*, Joseph L.-H. Tsui*, **Serina Chang***, Spyros Lytras, Mark P. Khurana, Samantha Vanderslott, Sumali Bajaj, Neil Scheidwasser, Jacob Liam Curran-Sebastian, ..., and Samir Bhatt
Nature 2025

[12] Measuring vaccination coverage and concerns of vaccine holdouts from web search logs
Serina Chang, Adam Fourney, and Eric Horvitz
Nature Communications 2024
Also presented at KDD 2023 Workshop on Epidemiology Meets Data Mining and Knowledge Discovery (oral); KDD 2023 Workshop on Data Science for Social Good (oral); IC^2S^2 2024 (oral)

[11] Inferring dynamic networks from marginals with iterative proportional fitting
Serina Chang*, Frederic Koehler*, Zhaonan Qu*, Jure Leskovec, and Johan Ugander
ICML 2024
Also presented at *Learning on Graphs* 2023

[10] Estimating geographic spillover effects of COVID-19 policies from large-scale mobility networks
Serina Chang, Damir Vrabac, Jure Leskovec, and Johan Ugander
AAAI 2023, AI for Social Impact Track
Also presented at KDD 2022 Workshop on Data-driven Humanitarian Mapping and Policymaking (oral); IC^2S^2 2022

[9] Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration
Dallas Card, **Serina Chang**, Chris Becker, Julia Mendelsohn, Rob Voigt, Leah Boustan, Ran Abramitzky, and Dan Jurafsky
Proceedings of the National Academy of Sciences (PNAS) 2022

[8] To recommend or not? A model-based comparison of item-matching processes
Serina Chang and Johan Ugander
ICWSM 2022
Also presented at IC^{2}S^2 2021 (oral)

[7] Data-driven real-time strategic placement of mobile vaccine distribution sites
Zakaria Mehrab, Mandy Wilson, **Serina Chang**, Galen Harrison, Bryan L. Lewis, Alex Tellionis, Justin Crowe, Dennis Kim, Scott Spillman, Kate Peters, Jure Leskovec, and Madhav Marathe
IAAI 2022

[6] Supporting COVID-19 policy response with large-scale mobility-based modeling
Serina Chang, Mandy Wilson, Bryan Lewis, Zakaria Mehrab, Komal K. Dudakiya, Emma Pierson, Pang Wei Koh, Jaline Gerardin, Beth Redbird, David Grusky, Madhav Marathe, and Jure Leskovec
KDD 2021, Applied Data Science Track
Best Paper Award (1 out of 705 submissions)

[5] Mobility network models of COVID-19 explain inequities and inform reopening
Serina Chang*, Emma Pierson*, Pang Wei Koh*, Jaline Gerardin, Beth Redbird, David Grusky, and Jure Leskovec
Nature 2021
Also presented at Networks 2021 (oral); NeurIPS 2020 COVID-19 Symposium (invited talk); NeurIPS 2020 Machine Learning for Health Workshop
Coverage in 650+ news outlets, including *The New York Times* and *The Washington Post*

[4] Epidemic dynamics in inhomogeneous populations and the role of superspreaders
Kyle Kawagoe*, Mark Rychnovsky*, **Serina Chang**, Greg Huber, Lucy M. Li, Jonathan Miller, Reuven Pnini, Boris Veytsman, and David Yllanes
Physical Review Research 2021

[3] Automatically inferring gender associations from language
Serina Chang and Kathleen McKeown
EMNLP 2019 (oral, short paper)

[2] Detecting gang-involved escalation on social media using context
Serina Chang, Ruiqi Zhong, Ethan Adams, Fei-Tzin Lee, Siddharth Varia, Desmond Patton, William Frey, Chris Kedzie, and Kathleen McKeown
EMNLP 2018 (oral, long paper)

[1] Crowd-sourced iterative annotation for narrative summarization corpora
Jessica Ouyang, **Serina Chang**, and Kathleen McKeown
EACL 2017 (oral, short paper)

INVITED TALKS

UCLA, Department of Communication, Colloquium Speaker Series	2027 (upcoming)
Public Health Association of Australia, AI & Public Health Conference **(Keynote)**	2026 (upcoming)
NeurIPS Workshop on LLMs for Social Reasoning and Simulation **(Keynote)**	2026 (upcoming)
INFORMS, Session on AI, Digital Platforms, and Human Attention	2026 (upcoming)
UC Berkeley, Cognition Colloquium (PSYCH 229A)	2026 (upcoming)
ACL Workshop on NLP and Computational Social Science, Panel on LLM Social Simulation	2026
MIT Workshop on AI and Elections	2026
Iowa State University, Translational AI Center Seminar Series	2026
Stanford University, Summer Institute in Computational Social Science (SICSS)	2026
Northwestern University, Validating Generative AI-Based Social Sciences	2026
University of Michigan, Program in Survey and Data Science, Likert Symposium	2026
Alameda County Public Health Department, All Epi meeting	2026
UC Berkeley, Berkeley AI Research (BAIR) Lab Seminar	2026
UC Berkeley, AI & Public Feedback Workshop	2026

UC Berkeley, Demography Brown Bag Series	2026
UC Berkeley, Institute of Transportation Studies Seminar	2026
UC Berkeley, Applied Statistical Methods (STAT 215B)	2026
UCSF, Deep Learning for Biological and Clinical Research (BMI/BioE 212)	2026
UVA Workshop on Social Contagions, Artificial Intelligence, and Democracy (**Keynote**)	2025
NeurIPS Workshop on LLM Persona Modeling (**Keynote**)	2025
Conference on AI and Statistics, ISU and National Institute of Statistical Sciences (**Plenary**)	2025
WHO Pandemic and Epidemic Intelligence Innovation Forum	2025
UC Berkeley, EECS Colloquium	2025
EPINEXT: Next-gen Methods for Data-Rich Epidemic Models (satellite of CCS2025)	2025
UC Berkeley, Biostatistics Seminar	2025
Frontiers in Computational Health, Berkeley, CA	2025
CHI 2025 Panels, “Opportunities and Challenges of Applying LLM-Simulated Data to HCI Studies”	2025
Princeton, Machine Behavior (COS 598B)	2025
Stanford, AI Agents and Simulations (CS 222)	2024
MIT, Initiative on Digital Economy (IDE) Seminar	2024
Learning on Graphs Conference, NYC Meetup	2024
UC Berkeley, Computational Precision Health Doctoral Seminar	2024
CMU, AI and Emerging Economies (Course 94-894)	2024
Cornell Tech, Urban Data (INFO 5430)	2024
Public Health Insight Podcast	2024
Oxford, Networks Seminar	2024
Stanford, Big Data Methods for Behavioral, Social, and Population Health Research (EPI 270)	2024
CMU School of Computer Science, Machine Learning Department	2024
UC Berkeley, EECS Department & Computational Precision Health	2024
University of Illinois Urbana-Champaign, Department of Computer Science	2024
AAAI’24 Workshop on Graphs and More Complex Structures for Learning and Reasoning	2024
University of Washington, Paul G. Allen School of Computer Science & Engineering	2024
Microsoft Research NYC, Computational Social Science	2024
Columbia Business School, Decision, Risk, and Operations Division	2024
Cornell, Departments of Computer Science & Information Science	2023
Johns Hopkins, Department of Computer Science	2023
MIT, Department of Political Science & Schwarzman College of Computing	2023
NYU Stern School of Business, Department of Technology, Operations, and Statistics	2023
Learning on Graphs Conference, Stanford Meetup	2023
Northeastern, Network Science Institute Seminar	2023
Cornell Tech, Applied Data Science: Decision-Making Beyond Prediction (ORIE 5355)	2023
Stanford, Graph Learning Workshop	2023
Columbia, Analysis of Networks and Crowds (COMS 6998)	2023
Cornell Tech, Data Science for Social Change (CS 6382)	2023
NSF Predictive Intelligence for Pandemic Prevention (PIPP), PandEval Research Team	2023
NIH National COVID Cohort Collaborative (N3C), Machine Learning Seminar	2023
Stanford, Algorithmic Fairness Seminar	2023
Stanford, Networks (MS&E 135)	2023
Meta, Computational Social Science Seminar	2023
Stanford, Introduction to Computational Social Science (MS&E 231)	2022
Stanford, Fundamental Concepts in MS&E (MS&E 302)	2022
OECD-ODISSEI Webinar on Open Data Infrastructure	2021
Data Science Connect Conference	2021
March for Science Podcast	2021
PathCheck Global Health Innovators Seminar	2021
Diaries of Social Data Research Podcast	2021
NeurIPS, COVID-19 Symposium	2020
AI Science Spotlight Series	2020
Global Pervasive Computational Epidemiology Seminar	2020
The Octavian Report Podcast	2020
Placekey Community Seminar	2020

Stanford, Stats ML Retreat 2020

TEACHING

Instructor, UC Berkeley, Machine Learning and Human Behavior (CS294-286) 2025-present
- This is a new graduate-level CS course that I created with projects, paper readings, and paper presentations
- Course material explores the intersection of ML and human behavior, including modeling human behaviors with ML, integrating human feedback into ML algorithms, and societal impacts of large-scale deployed ML systems (e.g., recommender systems)

Instructor, UCSF UC Berkeley, Public Health Practicum (CPH201B) 2026-present
- This is a graduate-level course for PhD students in the Computational Precision Health program
- Course material introduces students to public health research and practice, including lectures from public health experts and in-depth experiences where students conduct 1-1 interviews with public health professionals

Head course assistant (CA), Stanford University, Machine Learning with Graphs (CS224W) 2021
- Managed team of 9 CAs and class of around 300 students; oversaw lecture slides, assignments, exams, and final projects
- Course material covers the foundations and state-of-the-art of machine learning with graphs, including representation learning, graph neural networks, reasoning over knowledge graphs, and algorithms for large-scale networks

Instructor, Girls Who Code, Summer Immersion Program 2019
- Served as the primary teacher for a classroom of 20 high school girls
- Taught a 7-week curriculum including Python, HTML, CSS, JavaScript, and Arduino

Instructional assistant, Columbia University, Data Structures in Java (COMS 3134) 2017
- Led discussion sections, held weekly office hours, and graded assignments and exams

Peer tutor, Columbia University, Computer Science Theory (COMS 3261) 2017

ADVISING

Advising as Faculty, UC Berkeley

PhD:

Jessie (Zixin) Li, UCSF-UC Berkeley Computational Precision Health Program 2026-present
Joseph Suh, UC Berkeley EECS (co-advised with John Canny) 2024-present
Kelly (Yu Ying) Chiu, UC Berkeley EECS (co-advised with Stuart Russell) 2026-present
Zhining Zhang, UC Berkeley EECS 2026-present

Postdoc:

Bill (Wang) Zhu, UC Berkeley 2026-present

Undergraduate:

Ayush Raj, UC Berkeley 2025-present
David Zhai Yang, UC Berkeley 2025-present
Murari Ganesan, UC Berkeley → Snorkel.ai (Software Engineer) 2025-2026
Erfan Jahanparast, UC Berkeley → MIT CSAIL (Research Associate) 2024-2026
Steven Luo, UC Berkeley → CMU CS PhD 2025-2026
Jonathan Ngai, UC Berkeley → UC Berkeley Statistics MA 2025-2026

Visiting students:

Enock Niyonkuru, PhD student in UCSF BMI, rotation Dec 2025-Mar 2026
Zhiqing Hong, UC Berkeley visiting student, PhD student at Rutgers University Jul-Dec 2025

Research mentees during PhD, Stanford University
PhD: Alicja Chaszczewicz (Stanford CS), Jordan Troutman (Stanford CS)
Master's: Damir Vrabac (Stanford CS)
Undergraduate: Zhiyin Lin, Daisuke Masuda, Emma Wang, Benjamin Yan (Stanford); Maya Josifovska (UCLA)

SERVICE

Organization

Co-Founder and Co-Organizer, UC Berkeley Seminar on "Population Biology and Ecology of Infectious Diseases" 2026-present
Co-Organizer, Dagshtul Seminar on "Generative Agent-Based Simulation" 2027
Co-Organizer, KDD Workshop on Data-driven Decision Making for Public and Population Health 2026
Steering Committee, Center for Human-Compatible AI Workshop 2026
Co-Organizer, NeurIPS Workshop on Behavioral Machine Learning 2024
Co-Organizer, KDD Workshop on Epidemiology meets Data Mining and Knowledge discovery 2024

Program Chair, Machine Learning for Health (ML4H) 2023
Co-Organizer, KDD Workshop on Data Science for Social Good 2023
Panel Moderator, KDD Equity, Diversity & Inclusion (EDI) Day 2023
Organizer, NYC Digital Humanities Week, Using Computation to Analyze Gender in Film 2019

Journal reviewer

Science Advances 2026
Proceedings of the National Academy of Sciences (PNAS) 2026
Nature Communications 2026
BMC Public Health 2026
Proceedings of the National Academy of Sciences (PNAS) 2025
The New England Journal of Medicine AI (NEJM AI) 2024
Proceedings of the National Academy of Sciences (PNAS) 2024
Science Advances 2023
Nature Human Behaviour 2022
American Journal of Sociology 2021
ACM Transactions on Spatial Algorithms and Systems 2021
npj Urban Sustainability 2021

Conference reviewer

AAAI, Special Track on AI for Social Impact, Senior Program Committee 2027
Machine Learning for Health, Area Chair 2026
ACL Rolling Review 2026
KDD, Research Track, Area Chair 2026
IJCAI-ECAI, AI for Social Good Track, Senior Program Committee (Silver Tier Award) 2026
Machine Learning for Health, Area Chair 2025
AAAI, Special Track on AI for Social Impact, Senior Program Committee 2025
ACL Rolling Review 2025
UIST 2025
WWW 2025
AAAI 2025
Machine Learning for Health 2024
International Conference on Computational Social Science (IC^2S^2) 2024
KDD, epiDAMIK Workshop 2024
AAAI 2024
EMNLP 2024
Machine Learning for Health 2023
ACL Rolling Review 2023
ICWSM 2023
KDD, epiDAMIK Workshop 2023
KDD, Data Science for Social Good Workshop 2023
ACL Rolling Review 2022
Machine Learning for Health 2022
ACL, NLP for Positive Impact Workshop 2022
KDD, epiDAMIK Workshop 2022
ACL-IJCNLP 2021
Machine Learning for Health 2021
ACL, NLP for Positive Impact Workshop 2021
ICLR, AI for Public Health Workshop 2021

Mentorship

Machine Learning for Health (ML4H), Career Mentorship Program 2025
IJCNLP-AACL Student Research Workshop, Mentor 2025
International Conference for Computational Social Science (IC^2S^2), Mentor 2024
Stanford Computer Science, CURIS (undergraduate summer research), Mentor 2023
Machine Learning for Health (ML4H), Career Mentorship Program 2023
Stanford Computer Science, Undergraduate Mentoring Program 2022
Stanford Engineering, Summer Undergraduate Research Fellowship, Mentor 2021

WORK EXPERIENCE

AI Advisor, United Nations Development Programme (UNDP) 2025-present
- Serving as AI Advisor on "Project Vision: AI Unveils Tomorrow", which seeks to use AI to analyze how countries are responding to impending risks (e.g., climate change, pandemics, democratic backsliding) and adapting their development strategies

Consultant, Honda Research Institute USA 2025
- Advised on AI-related research projects

Research Intern, Microsoft Research 2022-2024
- Developed graph ML methods to mine search logs and derive public health insights. See Chang, Fourney, and Horvitz, "Measuring vaccination coverage and concerns of vaccine holdouts from web search logs", *Nature Communications* (2024).

Software Engineering Intern, Google, Geo Assistant 2018
- Built a new, user-facing feature for Google Search and Assistant; implemented changes from backend to frontend

Engineering Practicum Intern, Google, Search Site Reliability Engineering (SRE) 2017
- Improved internal tools for monitoring and tracking requests to Google Now